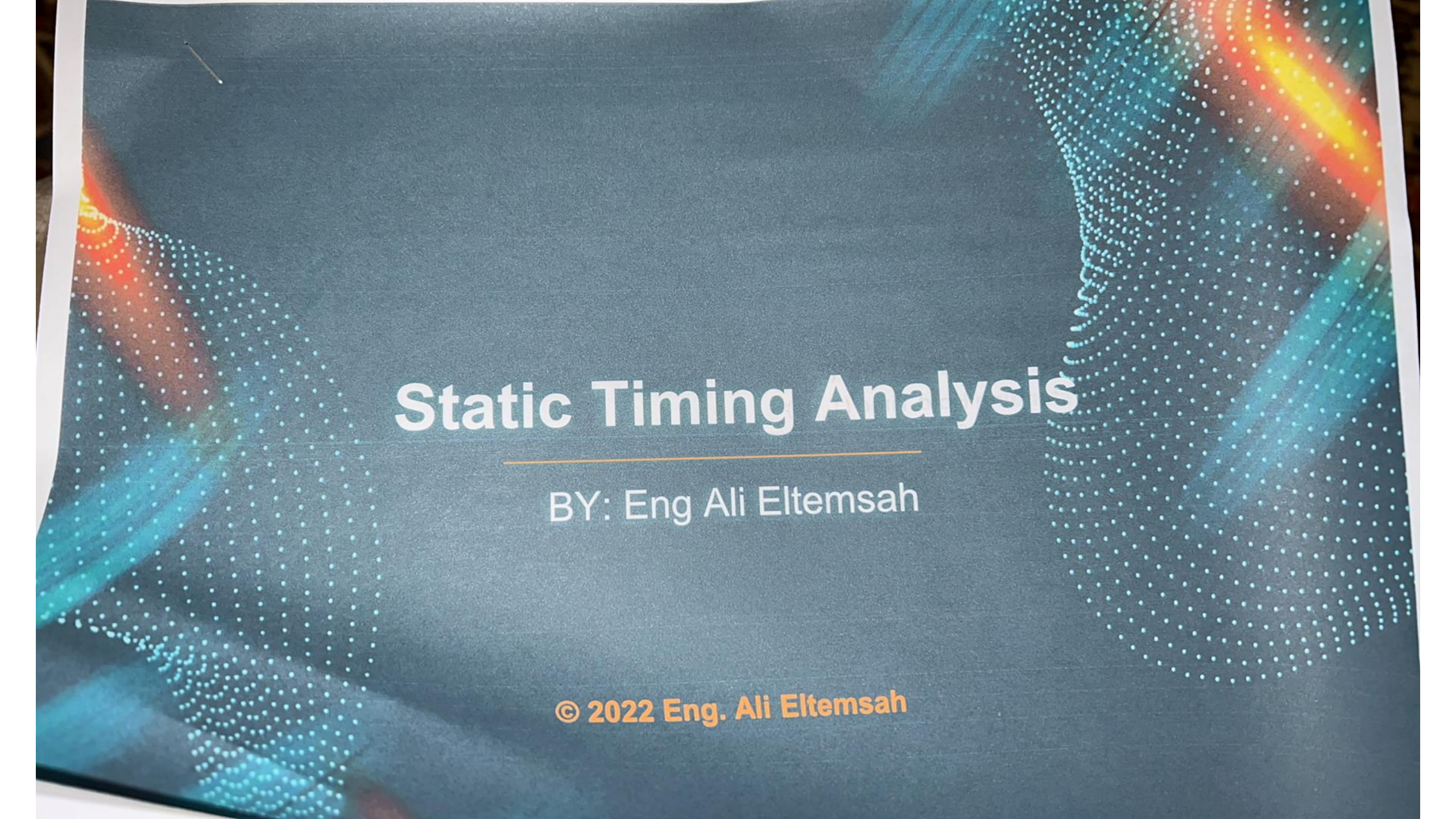




Static Timing Analysis

BY: Eng Ali Eltemsah

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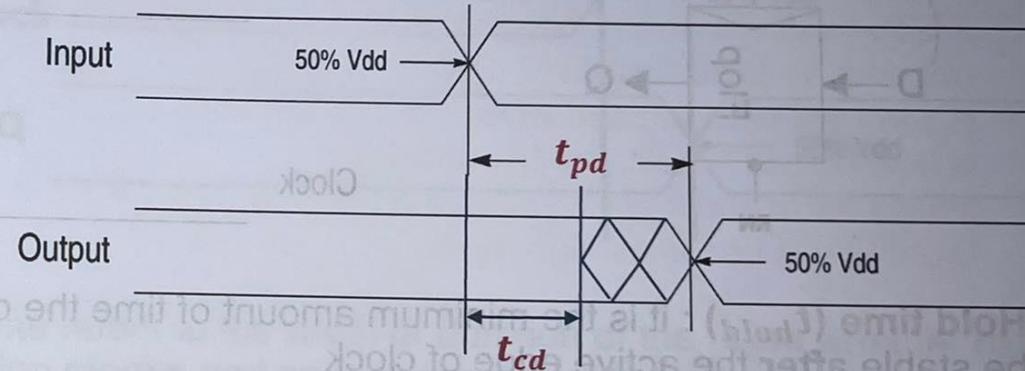
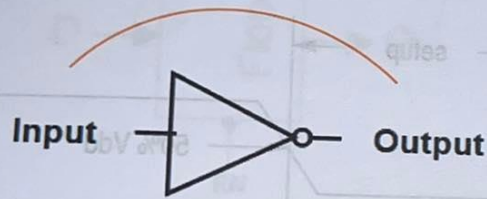
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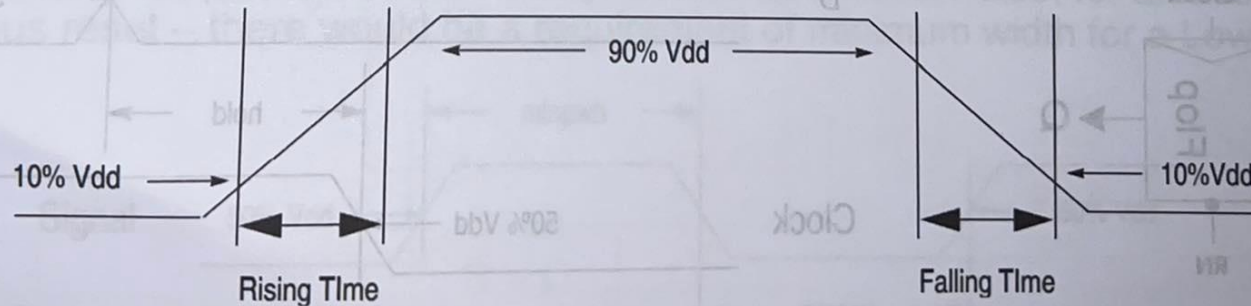
Understanding Combinational Arcs

Contamination Delay (t_{cd}) : The contamination delay through a cell is the **Minimum** time interval from when an input change until the output start to change its value.

Propagation Delay (t_{pd}) : The propagation delay through a cell is the **Maximum** time interval from when an input change until the output reach its **final** value.



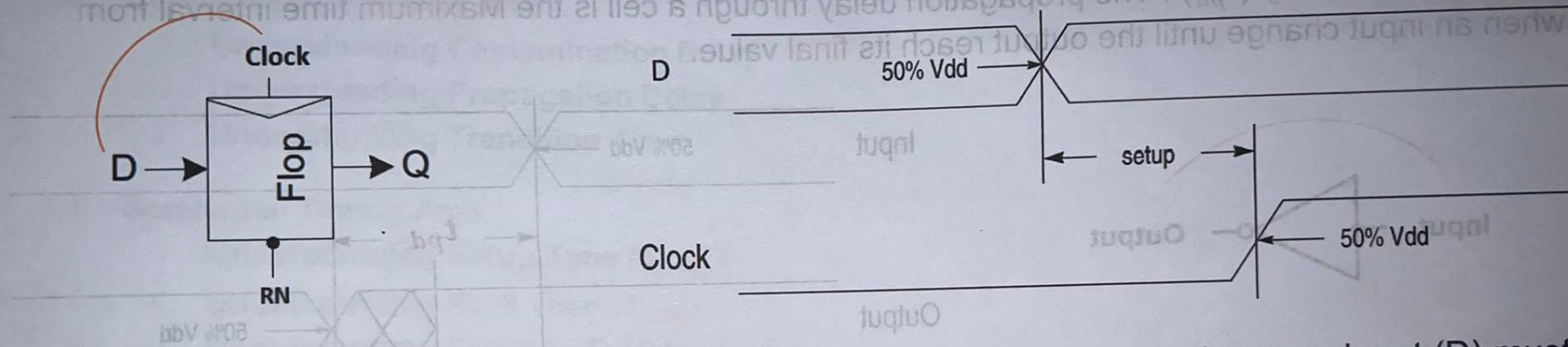
Transition Time (t_{rise} / t_{fall}) : The transition times (slews) on input and output pins are defined as the time interval between the signal crossing 10% of Vdd and 90% of Vdd.



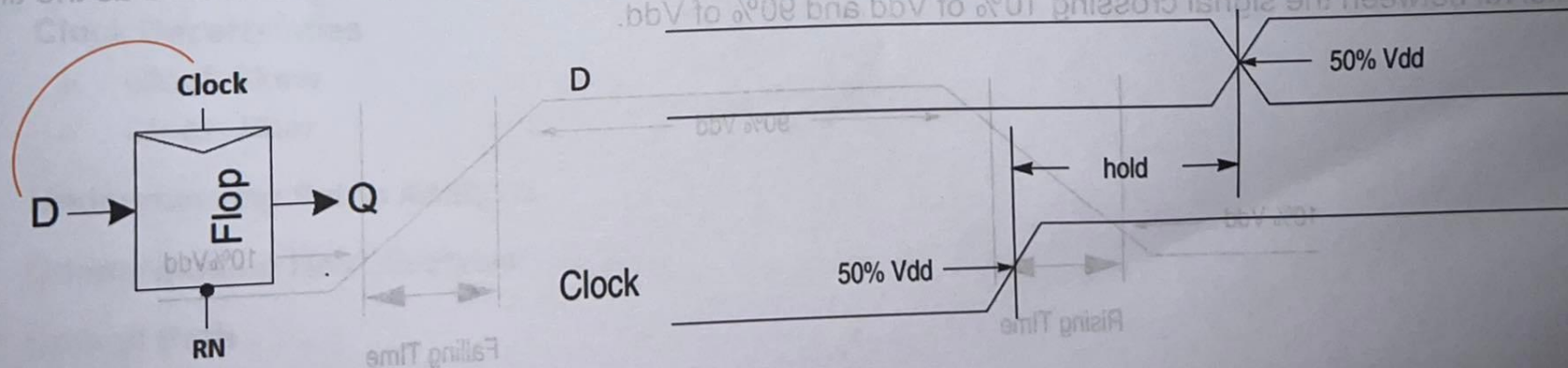


Understanding Sequential Arcs

- **Setup time (t_{setup})** : it is the minimum amount of time the data at the synchronous input (D) must be stable **before** the active edge of clock.

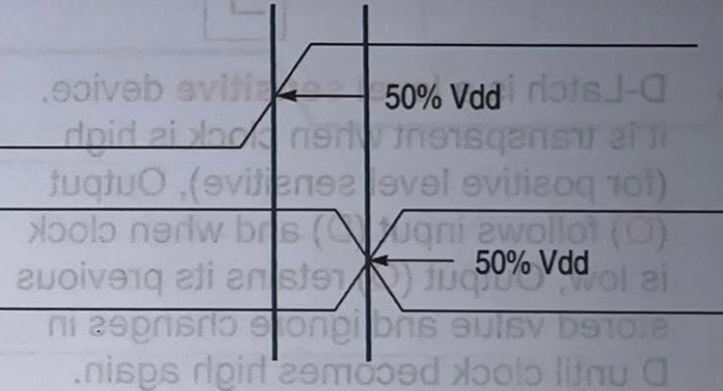
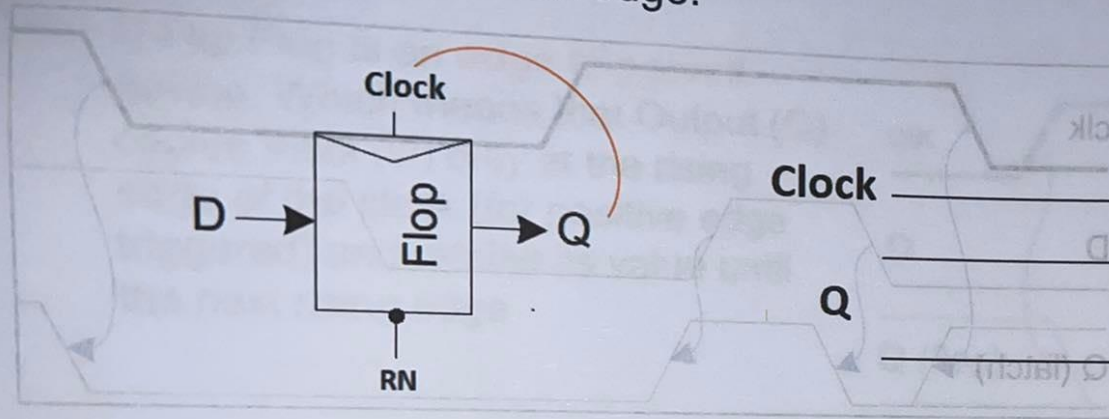


- **Hold time (t_{hold})** : it is the minimum amount of time the data at the synchronous input (D) must be stable **after** the active edge of clock

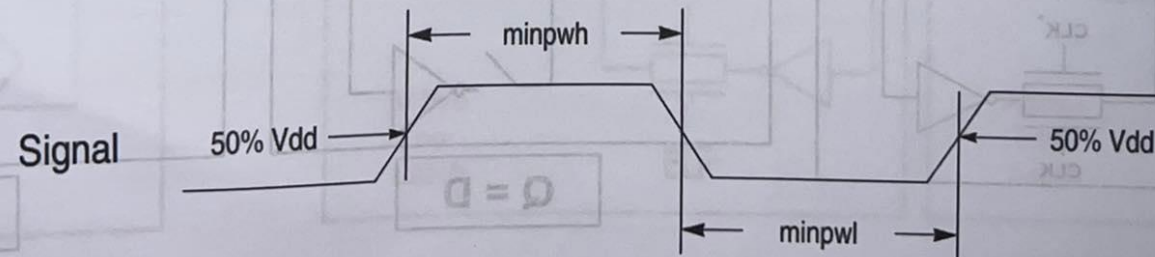


Standing Sequential Arcs

Clock-To-Q time (t_{c2q}) : it is the minimum amount of time for the flop output to be in a stable state after active clock edge.

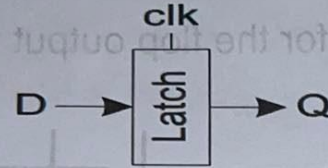


- **Minimum Pulse Width** : it is the time refers to the width or duration of the pulse whether high or low pulse, this check related to flop **clocks** and **asynchronous set/reset** pins. ☰
- For a clock pin, this check might refer to both Low Pulse as well as High Pulse.
- For an asynchronous pin, this refers to only the assertion level Pulse, for an active Low asynchronous reset – there would be a requirement of minimum width for a Low Pulse.

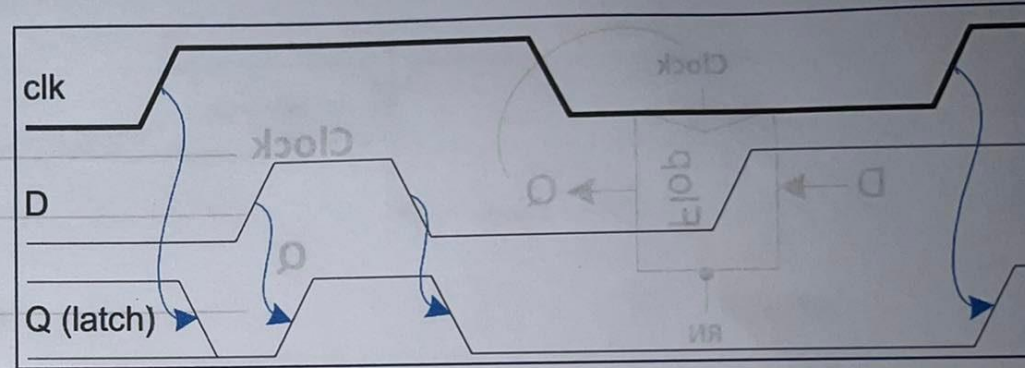


Latch

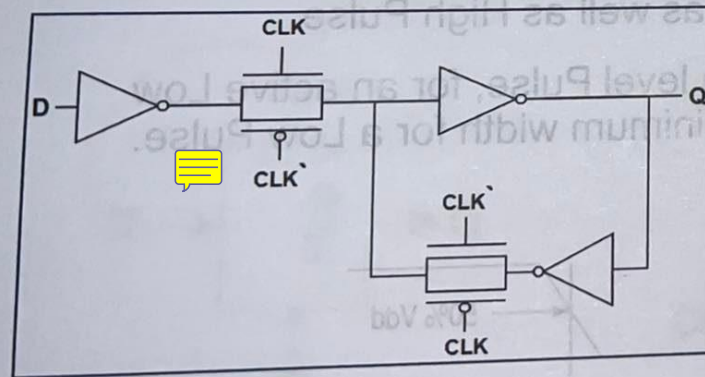
- D Latch



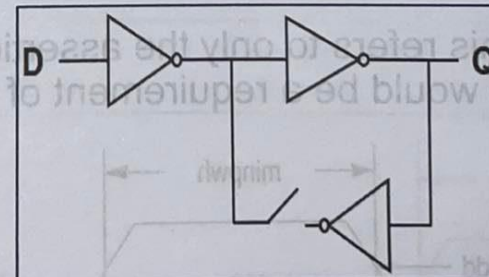
- D-Latch is a **level sensitive** device. it is transparent when clock is high (for positive level sensitive), Output (**Q**) follows input (**D**) and when clock is low, Output (**Q**) retains its previous stored value and ignore changes in D until clock becomes high again.



- D Latch Structure

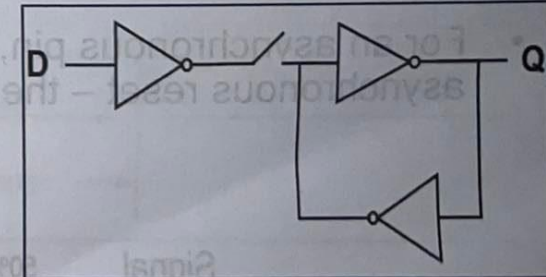


CLK = 1



Q = D

CLK = 0

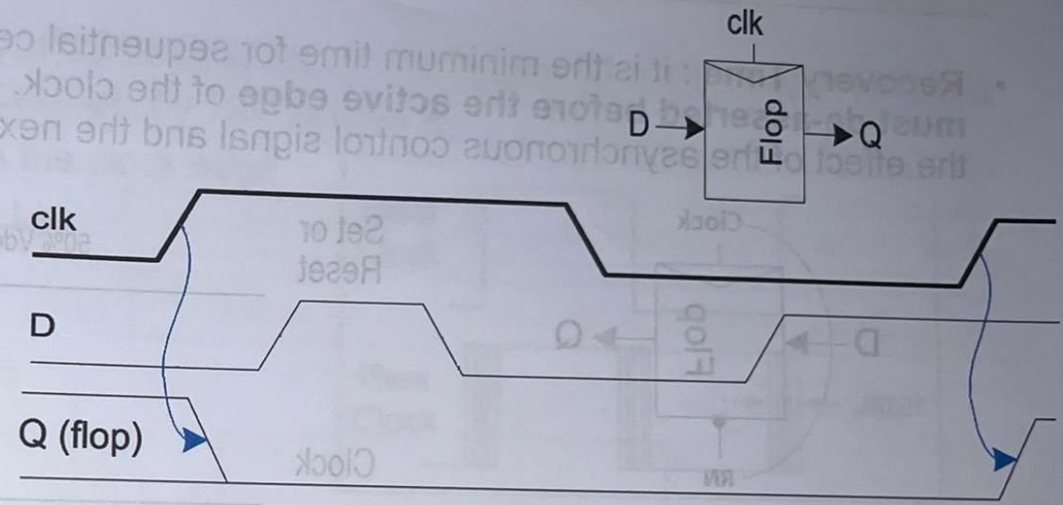


Q = Q

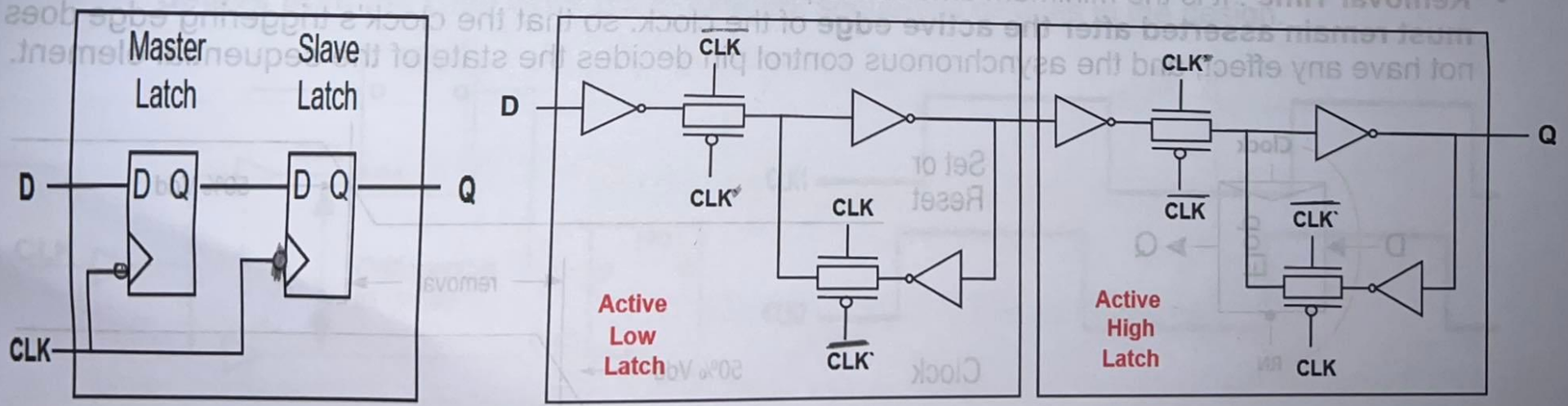
Flop

D Flip Flop

- D Flip Flop is an **edge triggered** device. Which means that Output (Q) copies input (D) only at the rising edge of the clock (for positive edge triggered) and retains its value until the next rising edge

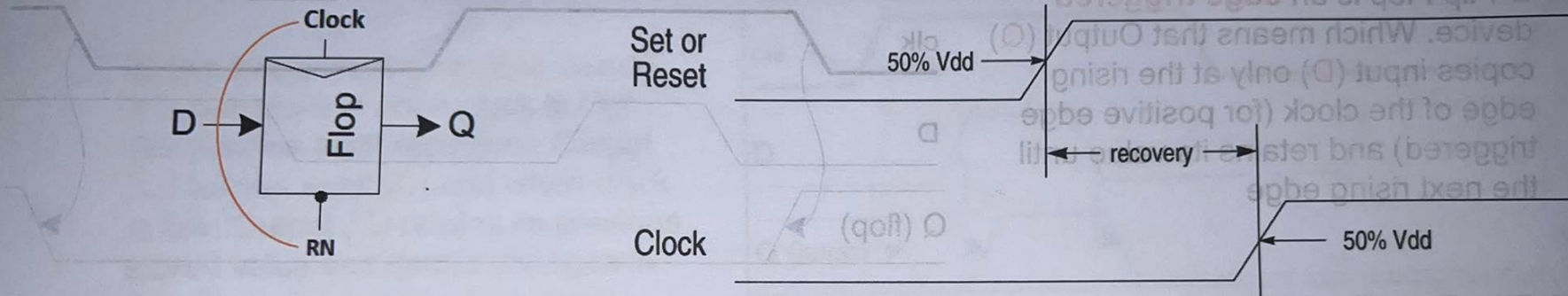


D Flip Flop Structure

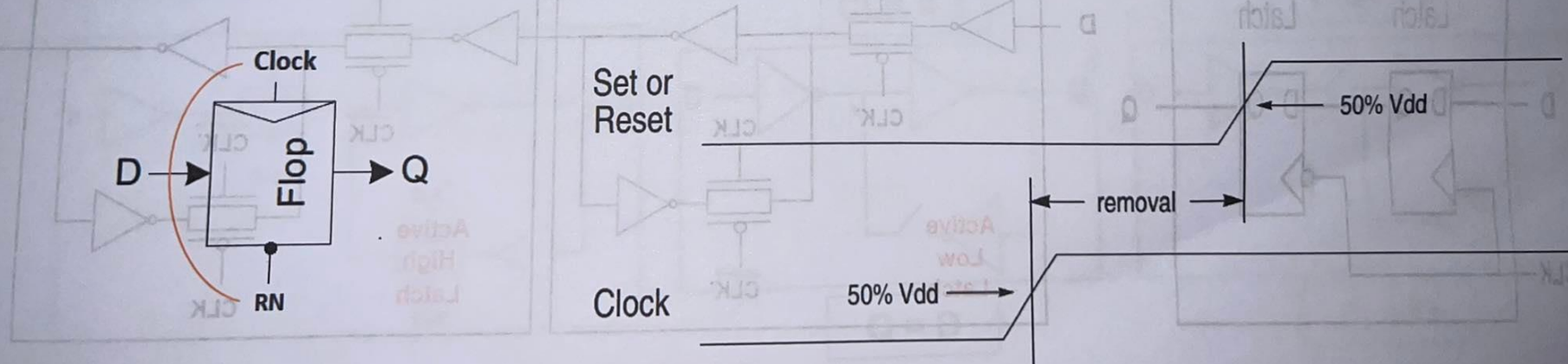


Understanding Sequential Arcs

- **Recovery Time** : it is the minimum time for sequential cells that the active low set or reset signal **must de-asserted before the active edge** of the clock. To allow the circuit to be recovered from the effect of the asynchronous control signal and the next clock edge can trigger the circuit



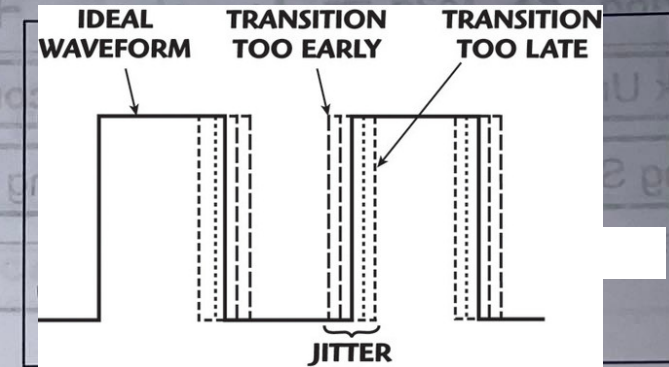
- **Removal Time** : it is the minimum time for sequential cells that the active low set or reset signal **must remain asserted after the active edge** of the clock. so that the clock's triggering edge does not have any effect, and the asynchronous control pin decides the state of the sequential element.



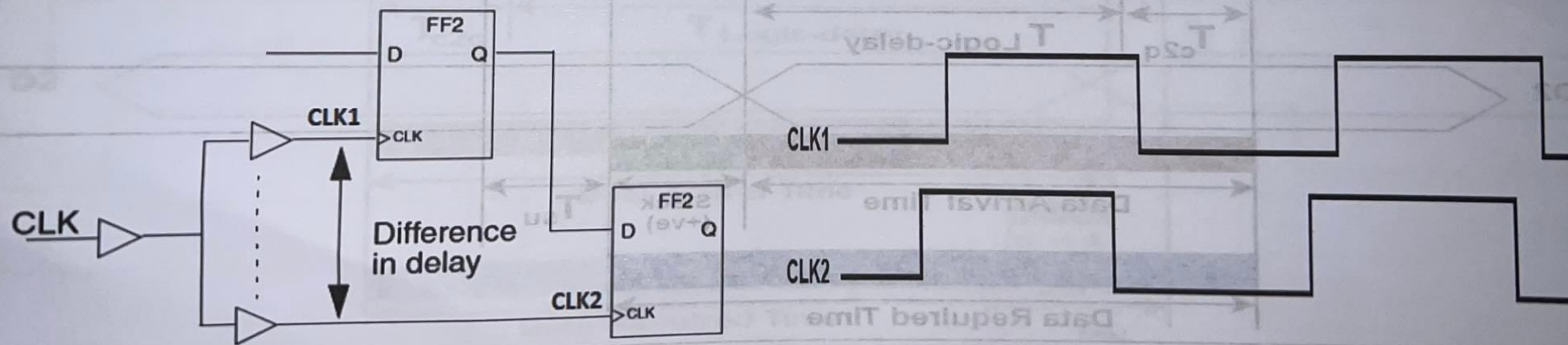
Understanding Clock Uncertainty

Clock Uncertainty : it is the time variation in successive edges of a clock with respect to the exact. Clock Uncertainty is caused by two reasons:-

1. **Clock Jitter**: it is the variation in the clock edge times of the clock source (ex: PLL) ☰



2. **Clock Skew**: it is the difference in the arrival times resulting from **different propagation delays** from the Chip clock pin to different sequential devices in the design. ☰



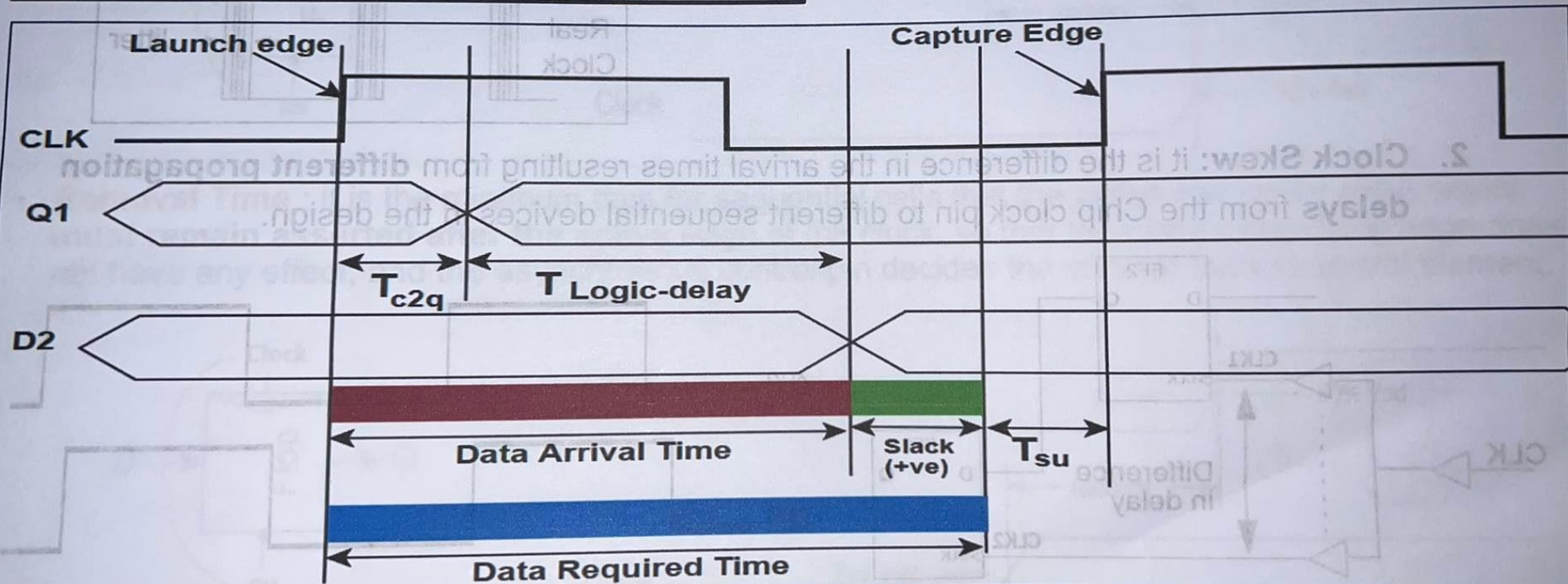
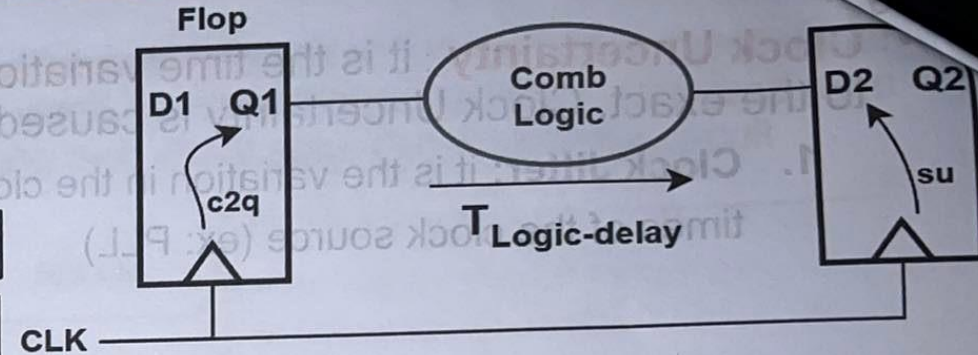
Understanding Setup Analysis

Setup Analysis Equation

$$T_{\text{period}} \gg T_{c2q} + T_{\text{Logic-delay}} + T_{\text{su}}$$

Clock Uncertainty (jitter+Skew) is **not considered**

Timing Slack is positive >> Setup Timing is met



Slack : it is the difference between data required time and data arrival time (Here slack is positive)

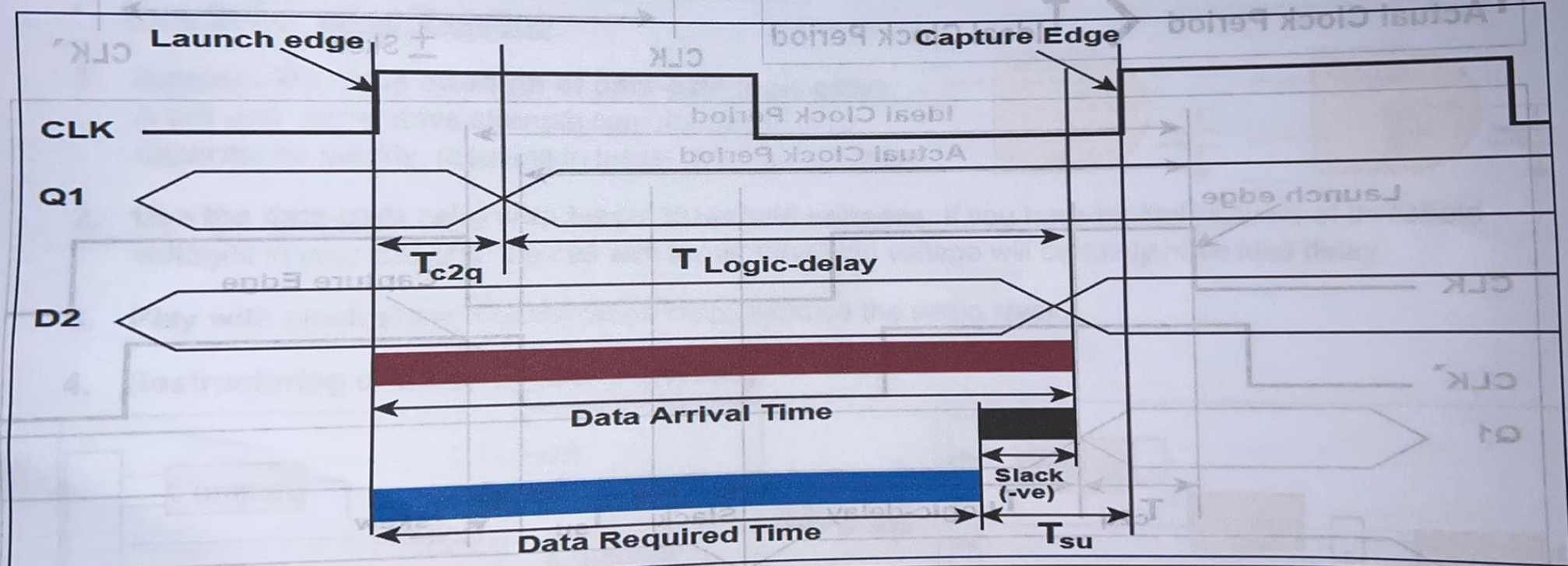
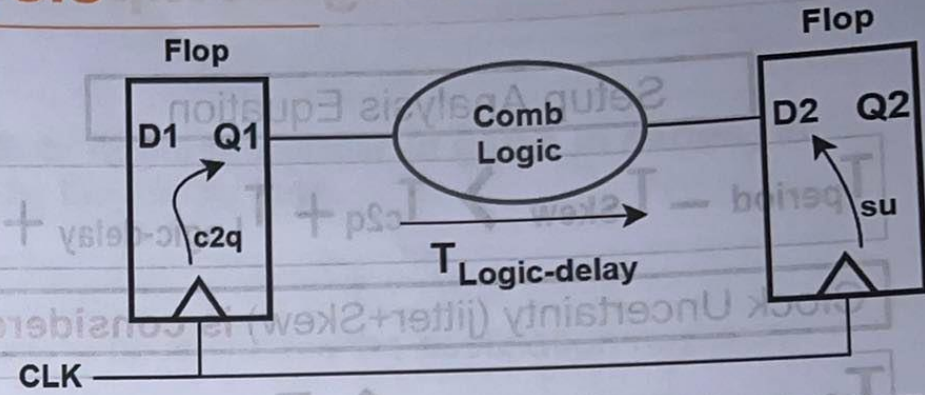
Understanding Setup Analysis

Setup Analysis Equation

$$T_{\text{period}} < T_{c2q} + T_{\text{Logic-delay}} + T_{\text{su}}$$

Clock Uncertainty (jitter+Skew) is **not considered**

Timing Slack is negative >> Setup Timing is Violated



Slack : it is the difference between data required time and data arrival time (Here slack is negative)

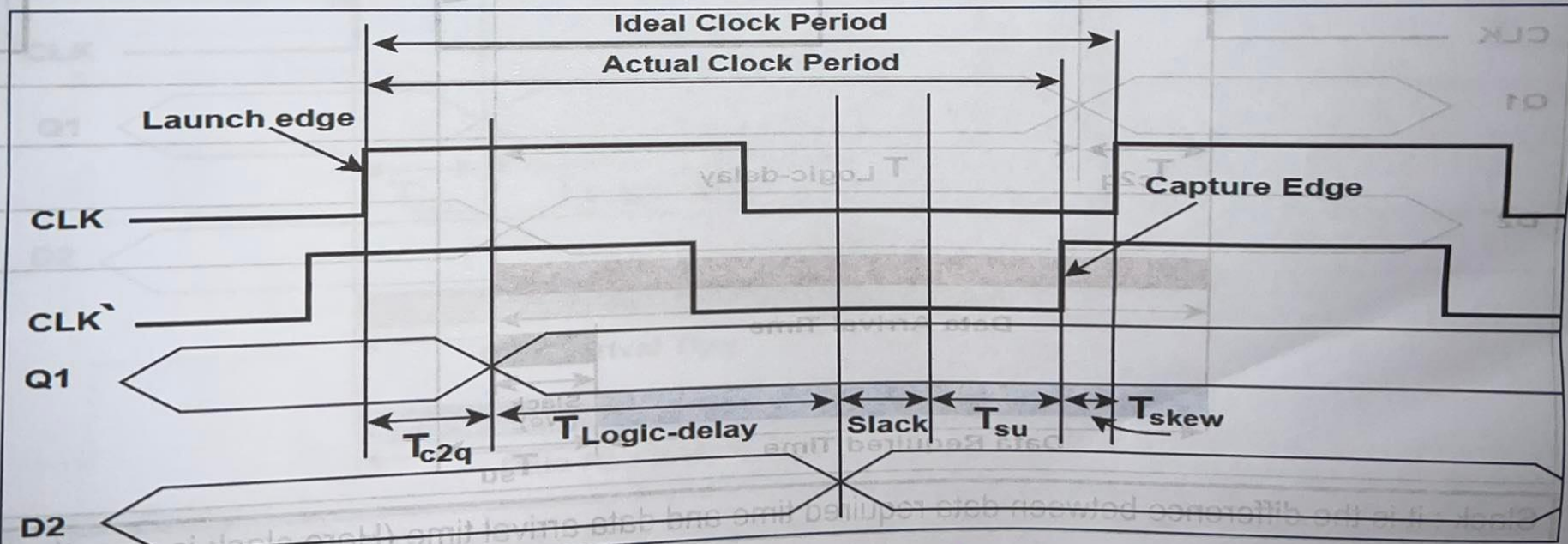
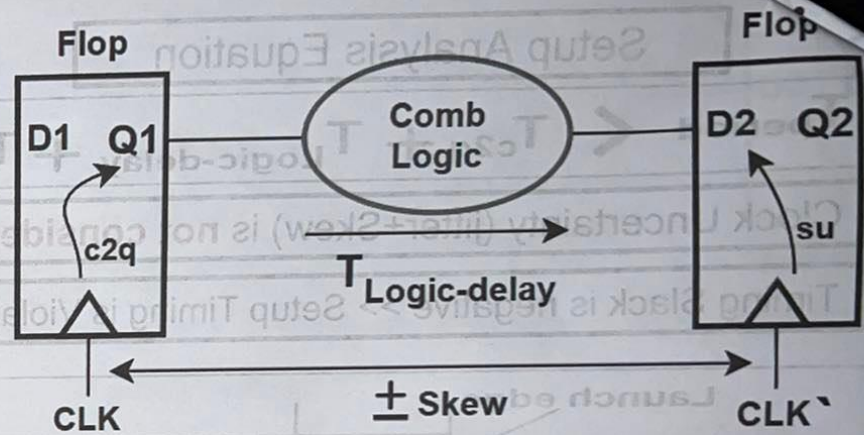
Understanding Setup Analysis

Setup Analysis Equation

$$T_{\text{period}} - T_{\text{Skew}} > T_{\text{c2q}} + T_{\text{Logic-delay}} + T_{\text{su}}$$

Clock Uncertainty (jitter+Skew) is considered

$T_{\text{Actual Clock Period}} < T_{\text{Ideal Clock Period}}$



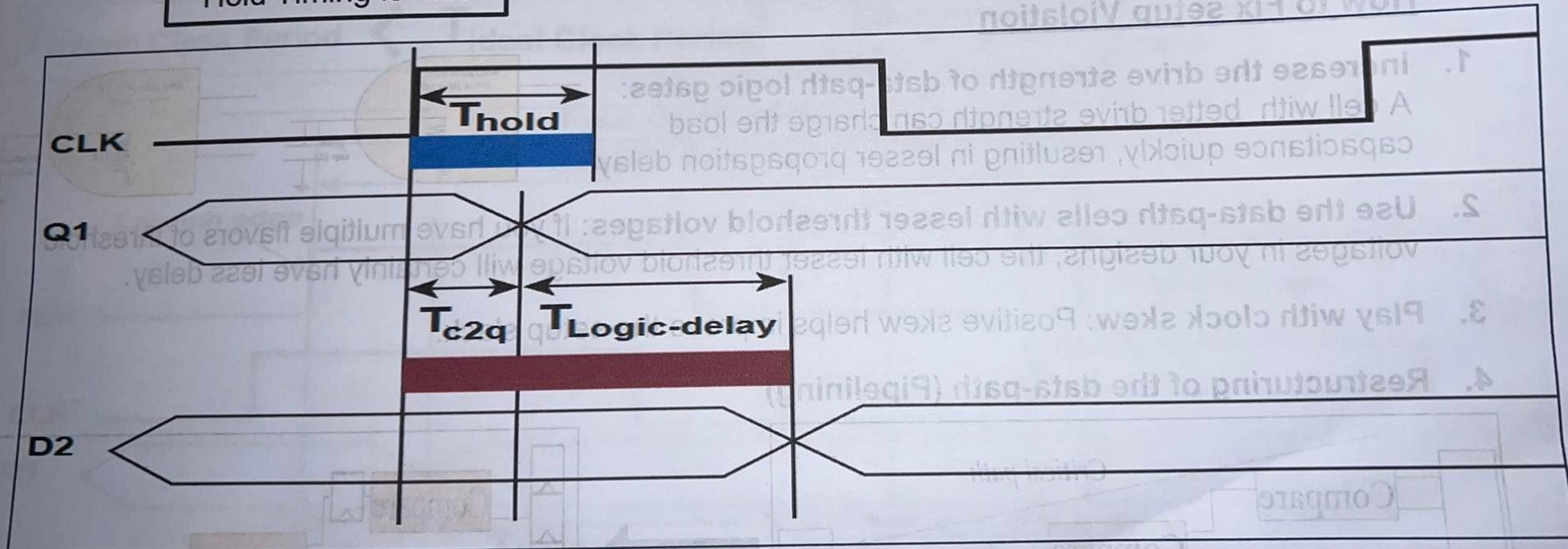
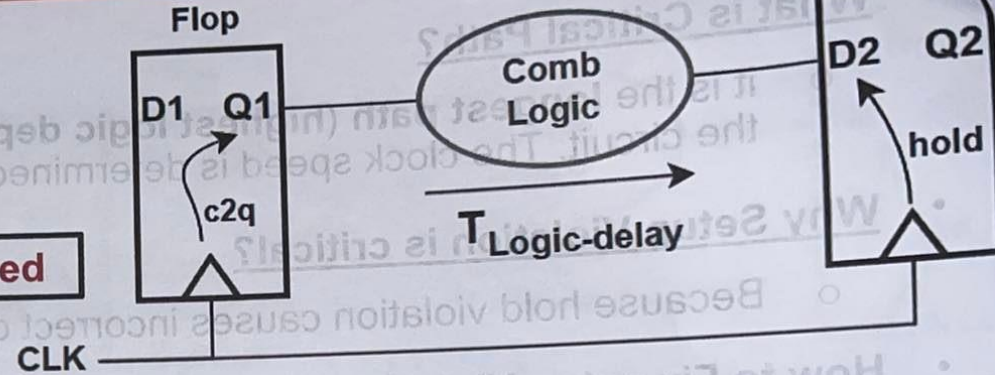
Understanding Hold Analysis

Hold Analysis Equation

$$T_{\text{hold}} < T_{\text{c2q}} + T_{\text{Logic-delay}}$$

Clock Uncertainty (jitter+Skew) is **not considered**

Hold Timing is met



This waveform represents that the data at D2 input **not changed** within the hold time (**No Hold Violation**)

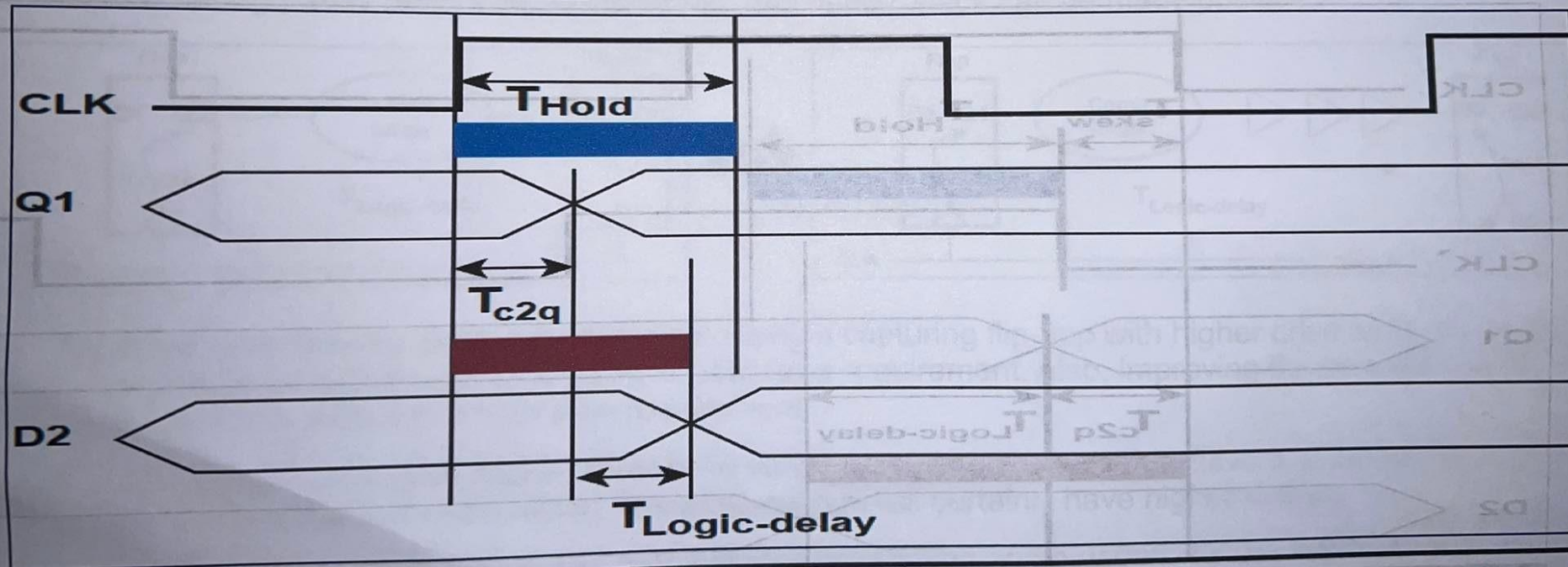
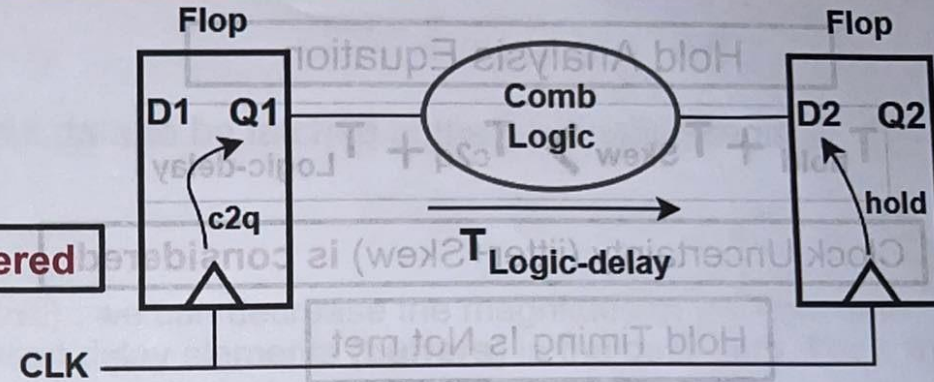
Understanding Hold Analysis

Hold Analysis Equation

$$T_{\text{hold}} > T_{\text{c2q}} + T_{\text{Logic-delay}}$$

Clock Uncertainty (jitter+Skew) is **not considered**

Hold Timing Is **Not met**



This waveform represent that the data at D2 input will is **changed** within the hold time (**Hold Violation**)

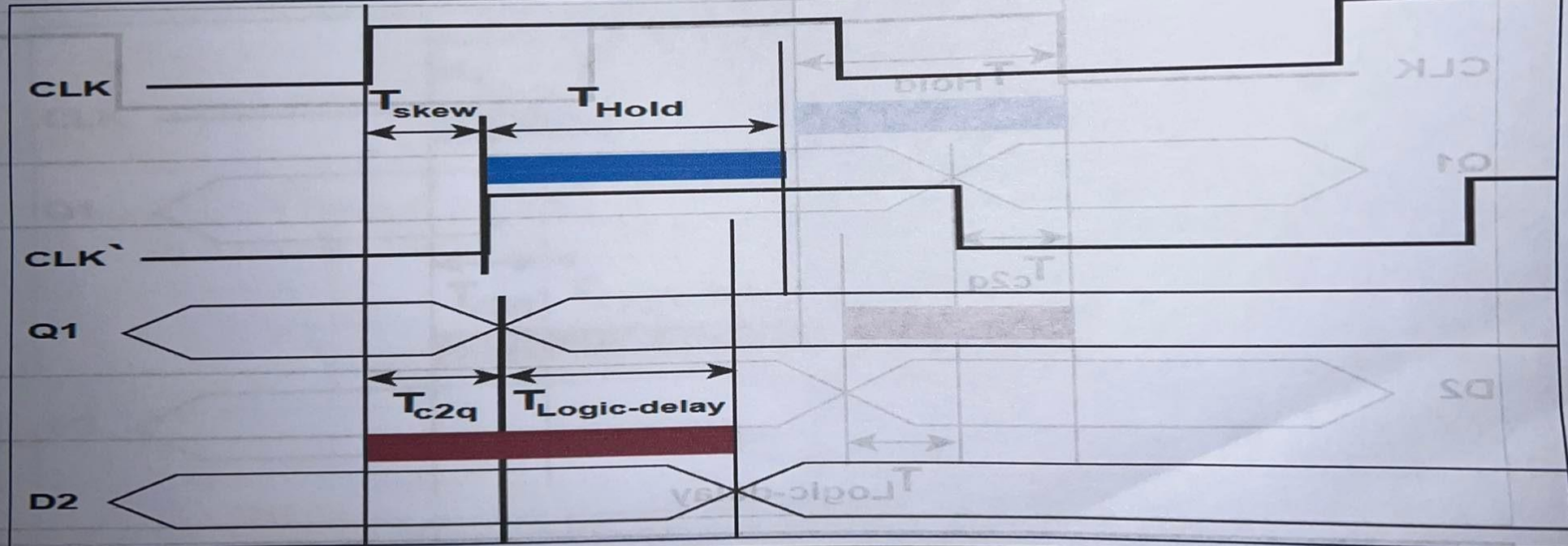
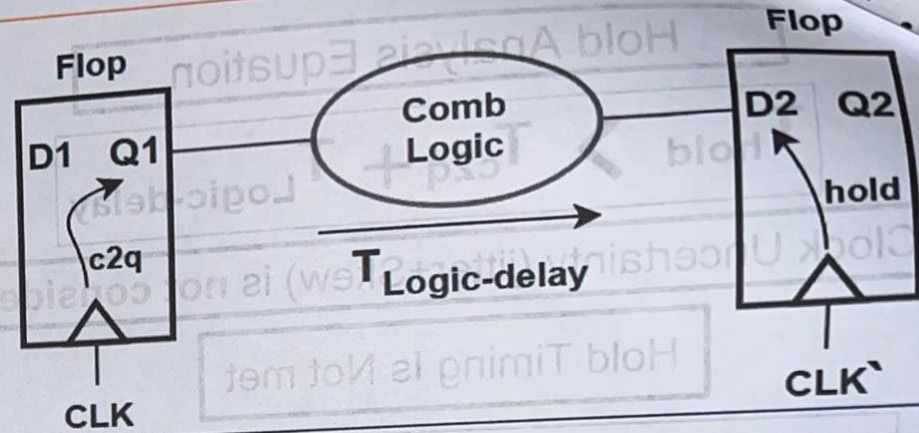
Understanding Hold Analysis

Hold Analysis Equation

$$T_{\text{hold}} + T_{\text{skew}} > T_{\text{c2q}} + T_{\text{Logic-delay}}$$

Clock Uncertainty (jitter+Skew) **is considered**

Hold Timing Is **Not met**



This waveform represents that the data at D2 input **is changed** within the hold time (**Hold Violation**)

Understanding Setup Analysis

What is Critical Path?

- Critical path is the longest path from a flip-flop output to a flip-flop input. The maximum operating frequency is determined by the critical path.

Why Setup Violation is critical?

- Because ^{Setup}hold violation causes incorrect data to be latched in the destination register.

How to Fix setup Violation

1. Increase the drive strength of data-path logic gates:

A cell with better drive strength can charge the load capacitance quickly, resulting in lesser propagation delay.



2. Use the data-path cells with lesser threshold voltages: If you have multiple flavors of threshold voltages in your designs, the cell with lesser threshold voltage will certainly have less delay.

3. Play with clock skew: Positive skew helps improve the setup slack.

4. Restructuring of the data-path (Pipelining)



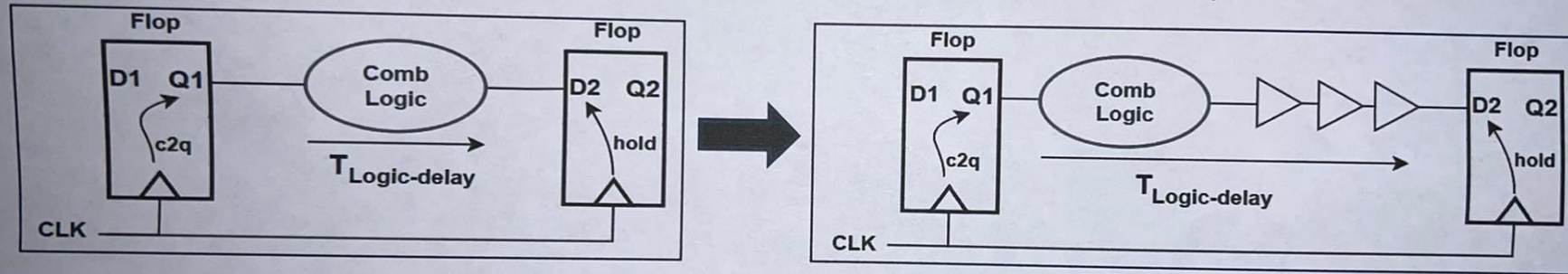
Understanding Hold Analysis

Why Hold Violation is critical?

- Because hold violation causes incorrect data to be latched in the destination register.

• How to Fix Hold Violation

1. **Insert delay elements (Most Commonly Used)** : we can decrease the magnitude of the hold time violation by increase in **data path delay** by insert delay elements (**buffers**) in the data-path. Thus, the hold violating path's delay can be increased, and hence, slack can be made positive.



2. **Improve hold time of capturing flip-flop:** Using a capturing flip-flop with higher drive strength and/or lower threshold voltage will give a **lower hold time** requirement. Also, improving the transition at flip-flop's clock pin reduces its hold time requirement.
3. **Use data-path cells with higher threshold voltages:** If you have multiple flavors of threshold voltages in your design, the cells with higher threshold voltage will certainly have higher delays.
4. **Play with clock skew:** A positive skew degrades hold timing and a negative skew aids hold timing.

